LEARNER GUIDE

Fractions, decimals and percentages fluently

Number Systems and Calculator Judgement · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

3/8, 0.375 and 37.5% are **one number** written three ways — one point on the number line. To change a fraction to a decimal, divide the numerator by the denominator; to change a decimal to a percentage, multiply by 100. Some fractions give decimals that stop (3/8 = 0.375); others give decimals that go on for ever (2/3 = 0.666…). To add or subtract fractions, cut them into pieces of the same size (a common denominator). Multiplying takes a fraction of a fraction, and dividing asks how many fit.

What you are learning to notice

- To compare numbers in different forms, write them all in the same form — or picture them on one number line.
- A fraction is in its simplest form when the numerator and the denominator have no common factor except 1.
- Dividing by a fraction asks how many of it fit: 3 ÷ 3/4 = 4.
- To find the whole from a part, find one part first: if 3/4 is 90, then 1/4 is 30.

Moving through it, one step at a time

For conversions, use the number line in your head: where does it sit? For calculations, start with a picture: a bar cut into equal pieces to add, an area to multiply, and 'how many fit?' to divide. The pictures show why the methods work: to add, use a common denominator; to multiply, multiply the numerators and multiply the denominators; to divide by a fraction, multiply by its reciprocal (the reciprocal of 3/4 is 4/3).

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Three names for one number

Write 0.35 as a fraction in its simplest form, and as a percentage.

1. 0.35 is 35 hundredths.	35/100
2. Divide the numerator and the denominator by 5.	7/20
3. Multiply the decimal by 100 for a percentage.	35%

0.35 = 7/20 = 35%.

Example 2 — Dividing by a fraction: from the picture to the method

Work out 2 1/2 ÷ 3/4.

1. Dividing asks how many 3/4 pieces fit into 2 1/2. Draw a bar and count.	3 pieces make 2 1/4; 1/3 of a piece more makes 2 1/2
2. So the answer is 3 and a third of a piece.	3 1/3
3. Why the method works: each whole holds 1 1/3 = 4/3 pieces of 3/4.	1 ÷ 3/4 = 4/3
4. So 2 1/2 wholes hold 2 1/2 × 4/3 pieces. Multiply.	5/2 × 4/3 = 20/6 = 10/3 = 3 1/3 ✓

2 1/2 ÷ 3/4 = 3 1/3. Dividing by 3/4 is the same as multiplying by its reciprocal, 4/3.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (equivalent fractions and adding fractions), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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